

Assignment 2

BCAC 0026 ANALYSIS OF ALGORITHM AND DATA STRUCTURES

Last date to submit assignment: 11-Feb-2025

Section A: Write the algorithm for

1. Push element in Stack
2. Pop element from Stack
3. Convert Infix to Postfix using Stack
4. Evaluate Postfix Expressions using Stack

Section B: Evaluate Postfix Expressions

Evaluate the following postfix expressions using a stack:

11. $5\ 3\ +\ 8\ *$
12. $6\ 2\ /\ 3\ 4\ *\ +$
13. $7\ 8\ +\ 3\ 2\ \wedge\ /$
14. $5\ 6\ 2\ +\ *\ 12\ 4\ /\ -$
15. $3\ 4\ 5\ *\ 6\ -\ +\ 7\ /$
16. $8\ 2\ \wedge\ 3\ *\ 4\ +$
17. $10\ 5\ /\ 6\ 3\ +\ *$
18. $15\ 7\ 1\ 1\ +\ -\ /\ 3\ *\ 2\ 1\ 1\ +\ +\ -$
19. $2\ 3\ 4\ +\ *\ 5\ /$
20. $9\ 3\ /\ 2\ 1\ +\ *$

Section C : Convert Infix to Postfix

Convert the following infix expressions to postfix notation:

1. $(A + B) * C$
2. $A + B * C$
3. $A * B + C / D$
4. $(A + B) * (C - D)$
5. $A + (B * C - (D / E \wedge F) * G) * H$
6. $(A + B) / (C - D) * E$
7. $A + B * (C \wedge D - E) / F$
8. $(A + B) * C - (D - E) / (F + G)$
9. $A * B + C * D - E / F$
10. $(A - B / C) * (D / E - F)$